

4	(a) the number of oscillations per unit time of the source/of a point on the wave/of a particle (in the medium) <i>or</i> the number of wavelengths/wavefronts per unit time passing a (fixed) point	M1 A1 (M1) (A1)	[2]
	(b) T or period = $2.5 \times 250 \text{ } (\mu\text{s})$ (= 625 μs)	M1	
	frequency = $1 / (6.25 \times 10^{-4})$ or $1 / (2.5 \times 250 \times 10^{-6}) = 1600 \text{ Hz}$	A1	[2]
	(c) (i) for maximum frequency: $f_o = f_s v / (v - v_s)$		
	$1640 = (1600 \times 330) / (330 - v_s)$	C1	
	$v_s = 8(.0) \text{ m s}^{-1}$ (8.049 m s^{-1})	A1	[2]
	(ii) loudspeaker moving towards observer causes rise in/high<u>er</u> frequency loudspeaker moving away from observer causes fall in/low<u>er</u> frequency <i>or</i>	B1 B1	[2]
	<u>repeated</u> rise and fall/higher and then lower frequency caused by loudspeaker moving towards and away from observer	(M1) (A1)	

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